



復旦大學



上海数学中心  
SHANGHAI CENTER FOR  
MATHEMATICAL SCIENCES

# Workshop on $p$ -adic Geometry



**SCMS, Fudan University**

October 8-12, 2026

**Venue:**

Gu Lecture Hall, Shanghai Center for Mathematical Sciences,  
Jiangwan Campus, Fudan University

**Speakers:**

- Ian Gleason, National University of Singapore
- Pol van Hoften, Zhejiang University
- Yongquan Hu, Morningside Center for Mathematics, CAS
- Naoki Imai, University of Tokyo
- Andreas Mihatsch, Zhejiang University
- Yong Suk Moon, BIMS
- Zicheng Qian, Morningside Center for Mathematics, CAS
- Xu Shen, Morningside Center for Mathematics, CAS
- Yousheng Shi, Zhejiang University
- Haining Wang, SCMS, Fudan University

**Organizers:**

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## Schedule

| Time        | Oct. 9             | Oct. 10             | Oct. 11         |
|-------------|--------------------|---------------------|-----------------|
| 09:00–09:30 | Registration       |                     |                 |
| 09:30–10:30 | Xu Shen            | Naoki Imai          | Yongquan Hu     |
| 10:30–11:00 | Tea Break          | Tea Break           | Tea Break       |
| 11:00–12:00 | Ian Gleason        | Yong Suk Moon       | Zicheng Qian    |
| 12:00–14:00 | Lunch              | Lunch               | Lunch           |
| 14:00–15:00 | Andreas Mihatsch   | Pol van Hoften      | Free discussion |
| 15:00–15:30 | Tea Break & Photos | Tea Break           |                 |
| 15:30–16:30 | Yousheng Shi       | Haining Wang        |                 |
|             |                    | Banquet<br>(18: 00) |                 |

## Titles & Abstracts

**Speaker:** Ian Gleason (National University of Singapore)

**Title:** Towards a deeper level cohomological local model diagram

**Abstract:** The local model diagram relates the integral model of a Shimura variety, via smooth morphisms, to the so-called local model, whose geometry is much easier to understand. In this talk we discuss a weak, purely cohomological version of this diagram, which can be established by  $v$ -sheaf-theoretic methods alone, in contrast to the classical approach through the deformation theory of  $p$ -divisible groups. This weak version admits natural generalisations to level structures deeper than parahoric, and it is already strong enough to recover many of the cohomological consequences usually extracted from the full diagram. As an example, we obtain a "coherent-side" formula for the cohomology of Shimura varieties at pro-Iwahori level. All results are joint with Xiangqian Yang and Xinwen Zhu.

**Speaker:** Pol van Hoften (Zhejiang University)

**Title:** Igusa stacks and the Jacquet--Langlands correspondence

**Abstract:** Igusa stacks are  $p$ -adic analytic objects associated to Shimura varieties of abelian type which link the Shimura variety to Fargues-Scholze's  $\mathrm{Bun}_G$ , the moduli stack of  $G$ -bundles on the Fargues-Fontaine curve. I will state a conjecture about exotic isomorphisms between Igusa stacks for different Shimura varieties, which is closely related to the exotic Hecke correspondences of Helm, Tian-Xiao and Xiao-Zhu. I will give a proof of this conjecture under some assumptions, and explain cohomological consequences. This is joint work with Jack Sempliner.

**Speaker:** Yongquan Hu (Morningside Center for Mathematics, CAS)

**Title:** Finite-length question for locally analytic representations of  $D^\times$

**Abstract:** Let  $D$  be the quaternion division algebra over  $\mathbb{Q}_p$ . In this talk, I will discuss the finite-length problem for spaces of locally analytic vectors in admissible unitary Banach representations of  $D^\times$  admitting an infinitesimal character.

**Speaker:** Naoki Imai (University of Tokyo)

**Title:** Local Langlands correspondence for  $p$ -adic covering groups

**Abstract:** Fargues-Scholze constructed the semisimplified version of the local Langlands correspondence for  $p$ -adic reductive groups using the idea of geometrization. In this talk, we discuss its generalization to covering groups of  $p$ -adic reductive groups. This is based on joint work in progress with Tony Feng, Teruhisa Koshikawa and Yifei Zhao.

**Speaker:** Andreas Mihatsch (Zhejiang University)

**Title:** Intersection theory on profinite families of RZ spaces

**Abstract:** We introduce a notion of  $\pi$ -divisible  $O$ -module which interpolates between strict  $O_F$ -modules for increasingly ramified extensions  $F/\mathbb{Q}_p$  and minuscule local shtukas for  $\mathbb{F}_p[[t]]$ . Starting with a framing object, we prove that the attached RZ space functor is representable by a formal scheme over  $\mathrm{Spf}(O_F)$ . We show that intersection numbers defined by special cycles vary locally constantly in such families, and we use this to deduce the unitary arithmetic fundamental lemma in equal characteristic from the mixed characteristic case. These results are joint work with Sebastian Bartling.

**Speaker:** Yong Suk Moon (BIMSA)

**Title:** Relative  $(\varphi, \Gamma)$ -modules and  $p$ -adic differential equations

**Abstract:** Let  $X$  be an affinoid rigid analytic space over a  $p$ -adic field, equipped with a suitable étale map to a unit polydisk. We will discuss a formalism of imperfect relative period rings over  $X$ , which lie inside the corresponding perfect relative period rings constructed by Kedlaya-Liu. Then we will explain a relative version of the Fontaine-Cherbonnier-Colmez equivalence between  $p$ -adic local systems on  $X$  and étale  $(\varphi, \Gamma)$ -modules over such imperfect relative period rings. From this equivalence, we will present a construction of  $p$ -adic differential equations attached to de Rham local systems, carrying both geometric and arithmetic differential operators. This is a joint work with Hansheng Diao and Zijian Yao.

**Speaker:** Zicheng Qian (Morningside Center for Mathematics, CAS)

**Title:** On the comparison between  $L$ -invariants

**Abstract:** We survey the generalization of Breuil's  $L$ -invariants to  $GL_n$ , including Breuil-Schraen  $L$ -invariants generalizing Schraen's work in case and Breuil-Ding  $L$ -invariants generalizing Breuil-Ding's joint work in case  $GL_3(\mathbb{Q}_p)$ . Firstly, BS  $L$ -invariants are defined based on careful computation of locally analytic cohomology of locally analytic generalized Steinberg, which is further based on a "restriction from  $G$  to  $\mathfrak{g}, N(T)$ " method and combinatorial computations in the Lie algebra setting. Secondly, BD  $L$ -invariants are defined based on explicit finite length admissible locally analytic representations constructed in a previous joint work with Breuil. Thirdly, BS  $L$ -invariants and BD  $L$ -invariants are related to Fontaine-Mazur  $L$ -invariants simultaneously through the pro-étale complex of the Drinfeld space, generalizing Vanhaecke's thesis in the  $GL_2(\mathbb{Q}_p)$  case. Finally, a version of local-global compatibility holds by combining the above intertwining on the Drinfeld space, Scholze's functor and its compatibility with generalized Hecke eigenspaces (following Kegang Liu's thesis) and an adjunction type result in Breuil's  $Ext^1$  paper. This talk is based on several joint works with Arnaud Vanhaecke and Benchao Su along with several of my own works (partially by rewriting and splitting my previous work with corrections).

**Speaker:** Xu Shen (Morningside Center for Mathematics, CAS)

**Title:** Rigid analytic 1-motives and conjugate uniformization of abeloid varieties

**Abstract:** We will first briefly discuss the theory of rigid analytic 1-motives over a  $p$ -adic field. In particular, we will discuss the Neron-Ogg-Shafarevich criteria. Next, we will talk about conjugate uniformization of abeloid varieties. This is a type of  $p$ -adic uniformization initiated by Iovita–Morrow–Zaharescu in case of abelian varieties with good reduction. Our approach is based on Fargues' theory of  $p$ -divisible rigid analytic groups. Along the way, we construct  $p$ -divisible rigid analytic groups from general rigid analytic 1-motives. This is joint work with Khai-Hoan Nguyen-Dang and Heer Zhao.

**Speaker:** Yousheng Shi (Zhejiang University)

**Title:** Intertwining operators and functional equations of Eisenstein series

**Abstract:** In this talk, I will present an explicit functional equation of Eisenstein series with vertex level. Along the way, we establish explicit formulas for the intertwining operator acting on the corresponding Siegel-Weil sections. In contrast with the previous methods, all the computation uses only the theory of local quadratic/Hermitian forms, hence answering a question of Kitaoka. This talk is based on a joint work with Qiao He, Chao Li, Tonghai Yang and Baiqing Zhu.

**Speaker:** Haining Wang (SCMS, Fudan University)

**Title:** On optimal  $p$ -adic uniformization of unitary Shimura curves

**Abstract:** In this talk, I will review the history of  $p$ -adic uniformization of Shimura curves and present a recent result concerning the  $p$ -adic uniformization of Shimura curves of unitary type. This is joint work with Michael Rapoport.

## Accommodation

### Hotel

Fraser Place Wujiaochang Shanghai 上海五角场辉盛坊公寓酒店  
1258 Yinhang Road, Yangpu District, Shanghai (杨浦区殷行路 1258 号)

### Directions from Pudong Airport (PVG, 浦东机场)

Taxi: It takes about 50 minutes and costs around 180 RMB.

Subway: Take Line 2 and Line 10 to Xinjiangwancheng (新江湾城) Station, then walk for about 5 minutes.

### From Hongqiao Airport (SHA, 虹桥机场) / Hongqiao Railway Station(虹桥火车站)

Taxi: It takes about 50 minutes and costs around 100 RMB.

Subway: Take Line 10 to Xinjiangwancheng(新江湾城) Station, then walk for about 5 minutes.

## Dining information

### Dining hall on Campus:

Each participant will receive 15 meal vouchers worth RMB 10 each, which can be used at the north dining hall on October 9<sup>th</sup> - 11<sup>th</sup>. (The meal vouchers are valid at all service windows, **except** the following: the Halal counter on the 1<sup>st</sup> floor, the central island section on the 2<sup>nd</sup> floor, the pay-by-weight section and roast duck counter on the 3<sup>rd</sup> floor.)

### Restaurants in UFUN Mall (悠方) :

Located near Gate 3 of Jiangwan Campus (江湾校区), a 15-minute walk from SCMS or a 5-minute walk from the hotel.

### Restaurants in Wujiaochang (五角场) Area:

There are many fancy restaurants in the shopping malls in this area. It takes 15 minutes to get there from the campus via subway Line 10.

# Map

